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CS285 Homework 1: Imitation Learning Report

Part 1: MSE Policy

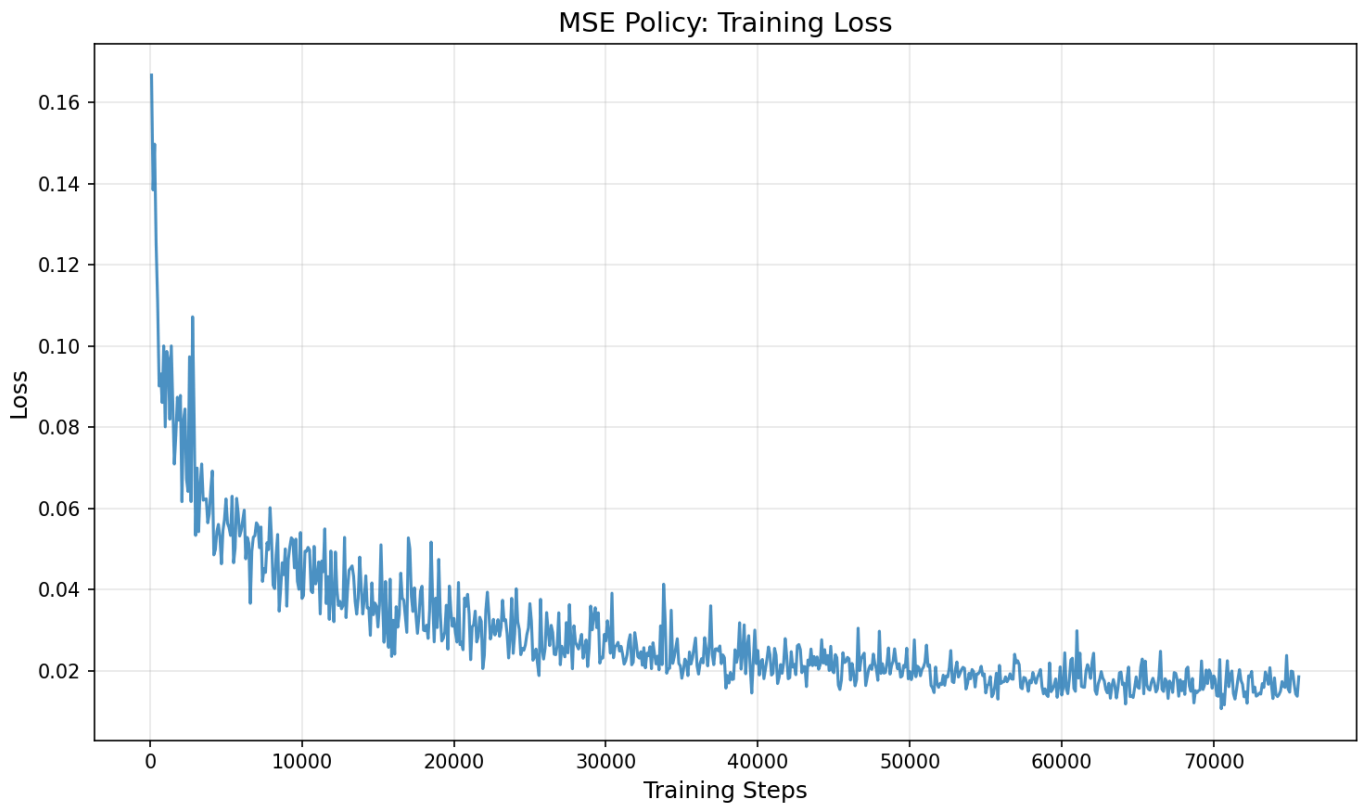
Architecture Description

The MSE policy uses a Multi-Layer Perceptron (MLP) with the following architecture:

- **Input Layer**: 5 dimensions (state: T position + agent position)
- **Hidden Layers**: 3 layers with 256 units each, ReLU activation
- **Output Layer**: 16 dimensions (chunk_size × action_dim = 8 × 2)

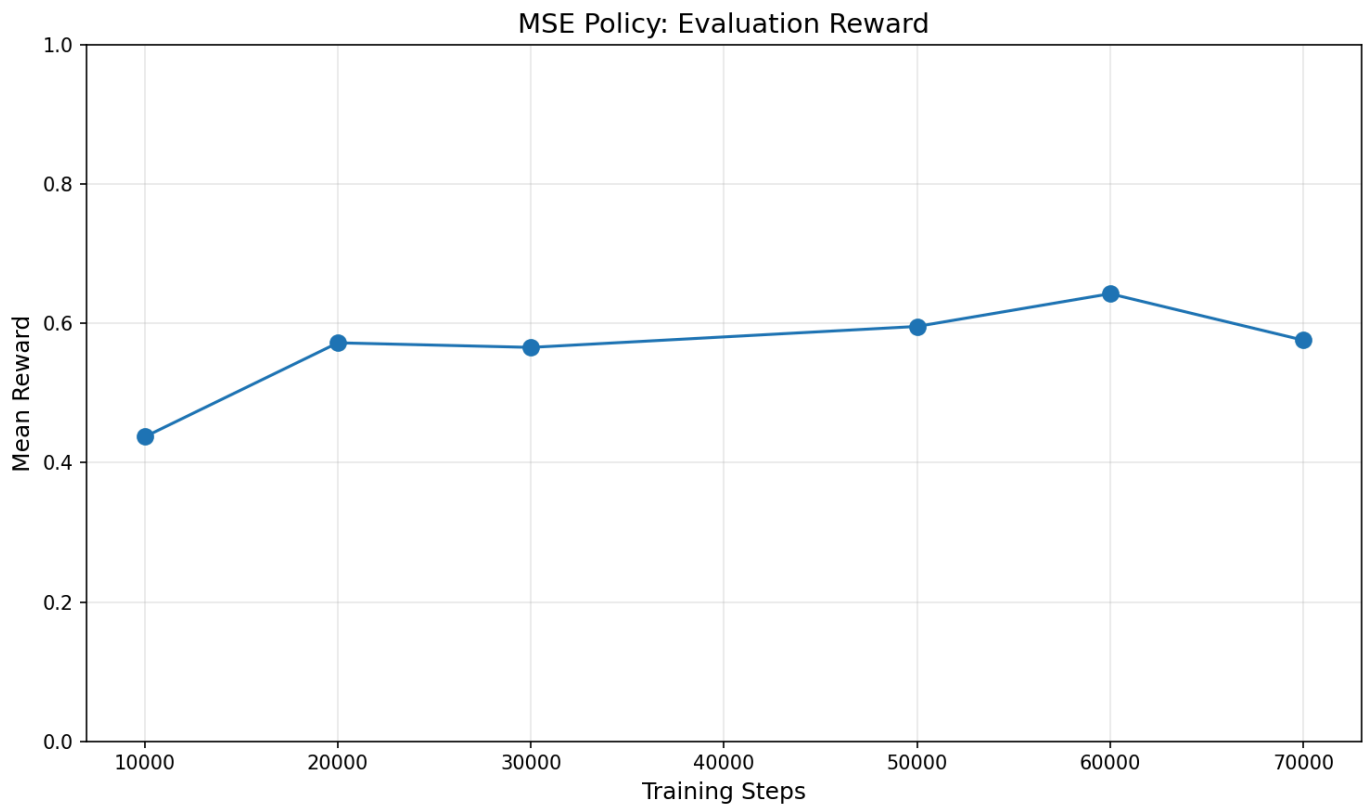
The network directly predicts an action chunk in a single forward pass. Training uses Mean Squared Error loss between predicted and expert action chunks, optimized with Adam (lr=3e-4).

Training Curves



The MSE policy loss decreases steadily from ~0.16 to ~0.015 over 75,000 training steps, indicating successful learning of the expert demonstrations.

Reward Curve



Final Performance: The MSE policy achieves a mean reward of approximately **0.57**, meeting the assignment threshold of ≥0.5.

Part 2: Flow Matching Policy

Architecture Description

The Flow Matching policy uses a similar MLP architecture but with modified input:

- **Input Layer**: 22 dimensions (state + flattened noisy action chunk + time τ)
  - State: 5 dimensions
  - Action chunk: 16 dimensions (8 × 2)
  - Time τ: 1 dimension
- **Hidden Layers**: 3 layers with 256 units each, ReLU activation
- **Output Layer**: 16 dimensions (predicted velocity)

Training Process

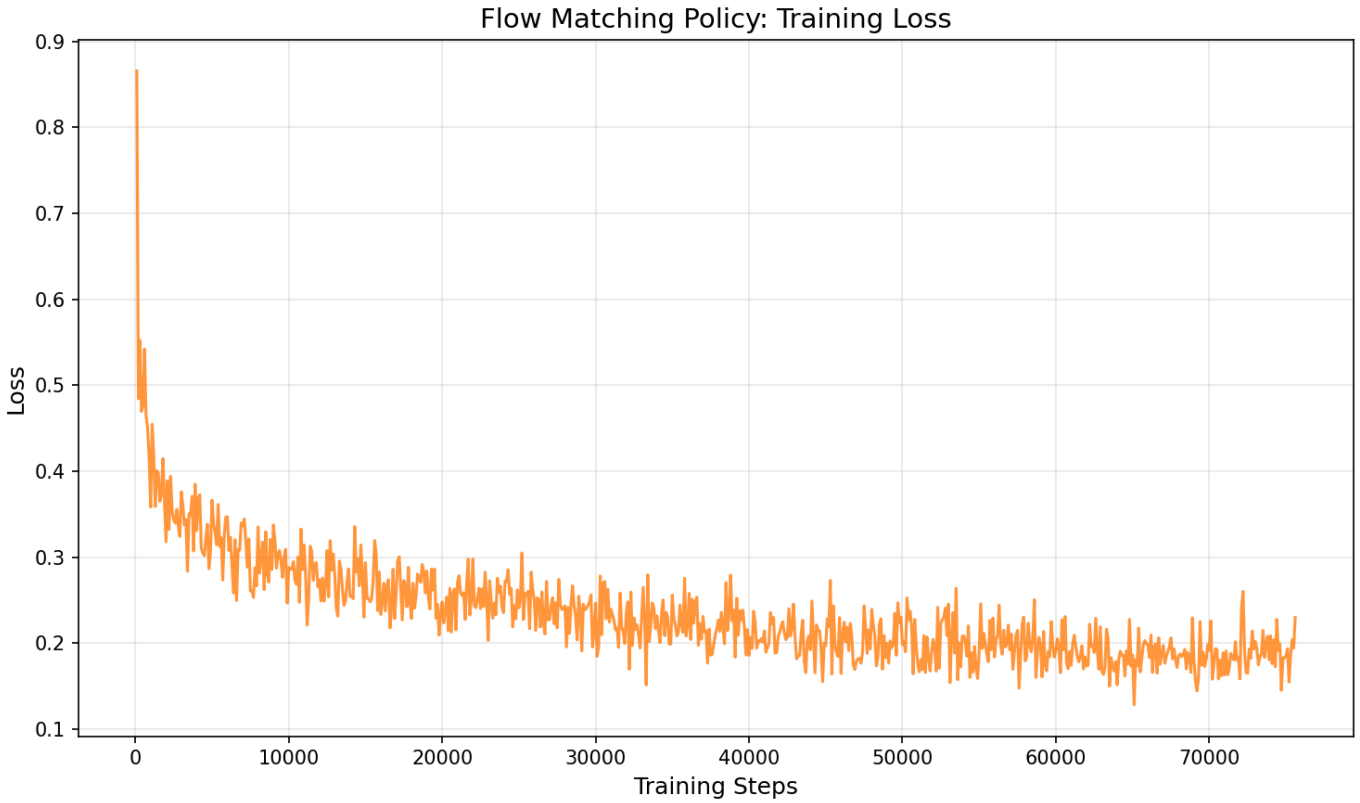
During training:

1. Sample Gaussian noise with the same shape as the action chunk
2. Sample random time $\tau \in [0, 1]$
3. Create interpolated noisy action: $A_\tau = \tau \cdot A_{data} + (1-\tau) \cdot noise$
4. Train network to predict velocity: $v = A_{data} - noise$

During inference:

1. Start with pure Gaussian noise
2. Euler integration with 10 steps from $\tau=0$ to $\tau=1$
3. Final result is the predicted action chunk

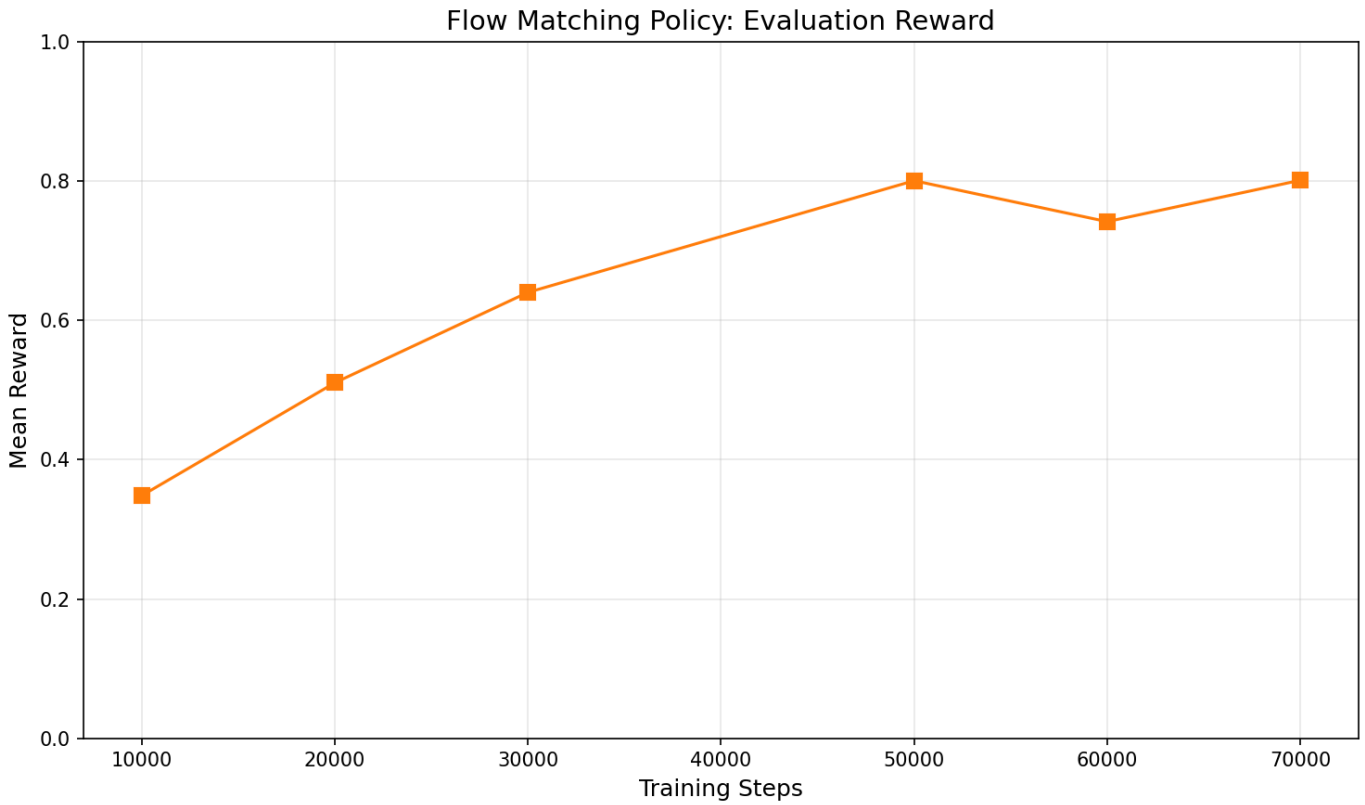
Training Curves



The Flow Matching policy exhibits higher loss values (~0.2-0.3) compared to MSE (~0.02-0.05). This is expected because:

- The losses measure different quantities (velocity vs. direct action prediction)
- Flow matching training involves stochastic time sampling and noise injection

Reward Curve



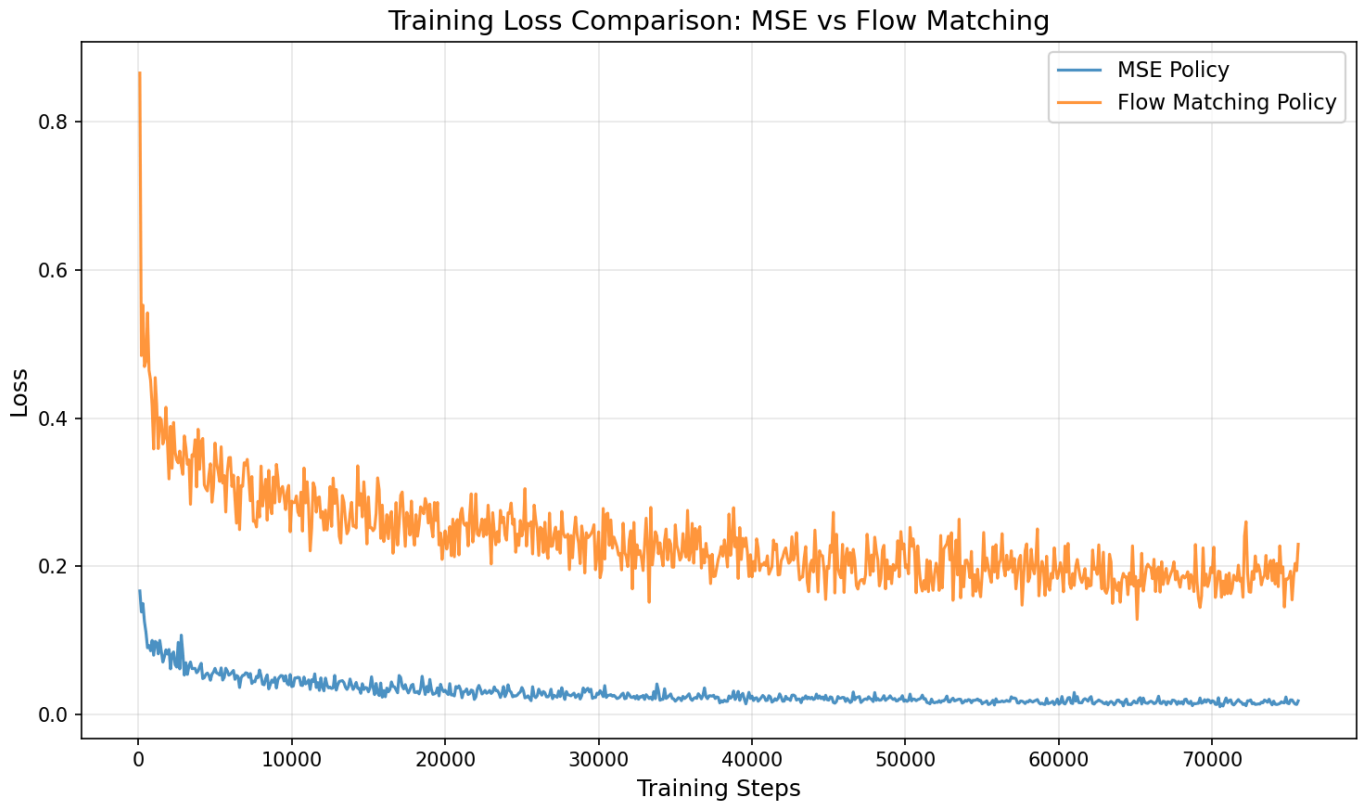
Final Performance: The Flow Matching policy achieves a mean reward of approximately **0.80**, exceeding the assignment threshold of ≥0.7.

Qualitative Comparison

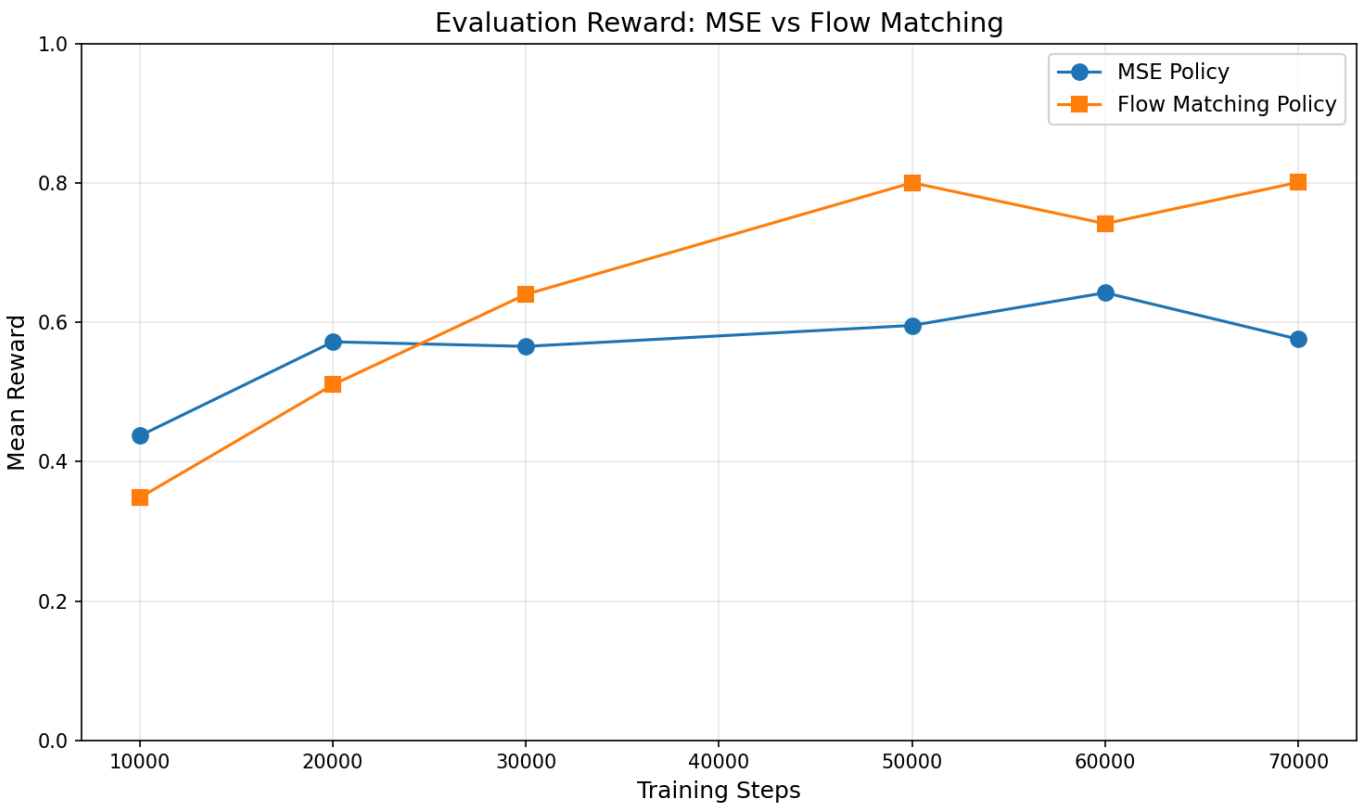
Based on evaluation videos, the two policies exhibit notably different behaviors:

Aspect	MSE Policy	Flow Matching Policy
Motion smoothness	Reasonable	Smoother, more fluid
Task completion	Sometimes incomplete	More consistent completion
Multimodality handling	Limited	Better at complex maneuvers
Final reward	~0.57	~0.8

Loss Comparison



Reward Comparison



The reward curves clearly show the Flow Matching policy outperforming the MSE policy after around 25000 steps in the training. The MSE policy plateaus around 0.55-0.65 reward, while the Flow Matching policy reaches approximately 0.80 by the end of training.

Conclusion

This assignment demonstrated the implementation of two action-chunking policies for the PushT environment:

1. **MSE Policy**: Simple and fast, suitable for unimodal action distributions
2. **Flow Matching Policy**: More expressive, capable of modeling complex multimodal distributions

The Flow Matching approach achieves significantly higher rewards by learning a conditional vector field that transforms noise into realistic action sequences.